

Advanced Practical Machine Learning

Project Rules & Guidelines

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Overview

This course is delivered as an intensive block over four consecutive days and culminates in a hands-on deep learning project. You will identify a real-world problem, source your own data, design and train a neural network in PyTorch, and deliver a working demo. The goal is not to follow a tutorial: it is to build something original that demonstrates your understanding.

Because the course runs over four consecutive days, you will scope and begin your project during the course and finalise it for submission a few weeks later (see Deliverables & Deadline). The live presentation has been replaced by a recorded project video, so you submit a polished demo and a clear walkthrough rather than presenting in the room.

Team Structure

- Teams of 2 students.
- Both members must contribute to all parts of the project (data, model, demo).
- In your submitted video, both team members must speak and explain parts of the code. Either member must be able to explain any part of the project.

Requirements

Your project must include all of the following:

1. **A real dataset.** You must find, collect, scrape, or access data from a real-world source. This is a core skill: you are data engineers.
2. **A deep learning model in PyTorch.** Your model must be a neural network (at least one hidden layer). You must write and understand the training loop. Using torch.nn modules is fine, but you must be able to explain every component.
3. **A working demo.** At minimum, a Streamlit application where a user can interact with your trained model (provide input, see predictions). Bonus for polished UI, visualizations, or deployment.
4. **A project video.** In place of a live presentation, you will record and submit a video that walks through your project, your code, and a live demonstration of your application. Both team members must appear and explain parts of the codebase, and each must be able to explain the entire project. Recommended length: 8 to 15 minutes.

What is Banned

Banned Datasets & Tasks

The following are not acceptable as your primary dataset or task:

- MNIST, Fashion-MNIST, CIFAR-10, CIFAR-100
- Titanic, Iris, Boston Housing, Wine Quality
- Any Kaggle “Getting Started” competition dataset
- Any dataset that comes pre-packaged with a tutorial you followed

Why? These are learning exercises, not projects. You have already moved beyond them.

Banned Approaches

- Decision trees, random forests, gradient boosting (XGBoost, LightGBM). These are not deep learning.
- Scikit-learn classifiers or regressors as your main model.
- Using a pre-trained model as-is with no fine-tuning or adaptation (simply calling a model from HuggingFace without any training is not a project).
- Copy-pasting an entire notebook or tutorial and changing the dataset. Your architecture and training code must be your own work.

What IS Allowed

- Transfer learning and fine-tuning pre-trained models (with clear explanation of what you changed and why).
- Using torchvision, torchaudio, torchtext for data loading and transforms.
- Using standard PyTorch modules (nn.Linear, nn.Conv2d, nn.LSTM, etc.).
- Combining deep learning with other techniques, as long as the neural network is the core of your project.
- Using external APIs or web scraping to build your dataset.

Evaluation Criteria

Criterion	Weight	What We Look For
Originality	15%	Is the problem interesting? Did you find a real dataset? Is this more than a tutorial copy?
Model Design	20%	Why this architecture? Can you justify your choices? Does the model match the problem?
Code Quality	15%	Clean, readable code. Proper training loop. Correct use of PyTorch.
Working Demo	15%	Does the Streamlit app work? Can a user interact with it? Is the demo clearly shown in the video?
Video Presentation & Defense	35%	Can you clearly explain every part of your code on camera? Do both members demonstrate understanding of the model and the choices behind it? If you cannot explain your code, nothing else matters.

The video replaces the live oral defense. It is weighted heavily because it is the main way you demonstrate that the work and the understanding are genuinely yours.

Project Ideas (Starting Points)

These are examples to inspire you: do not use them verbatim. Find your own angle:

- Classify satellite images of a specific region (e.g., detecting deforestation, urban growth)
- Predict energy consumption from weather and building sensor data
- Analyze medical imagery from an open clinical dataset (X-rays, dermatology)
- Generate or classify music genres from audio spectrograms
- Detect anomalies in network traffic or IoT sensor data
- Sentiment analysis on a niche domain (legal documents, scientific papers, product reviews in a specific industry)
- Time series forecasting with LSTMs on financial, environmental, or industrial data

Deliverables & Deadline

1. **A GitHub repository** with all code, the trained model, and the Streamlit demo.
2. **A project video** (recommended 8 to 15 minutes) covering the problem, the dataset, the model and training code, and a live walkthrough of the working demo. Both team members must speak.

The course runs over four consecutive days: [course dates]. You will scope and begin your project during the course. The GitHub repository and the video must both be submitted by [submission deadline], a few weeks after the course ends.

Note: replace [course dates] and [submission deadline] with your actual dates before sharing.

Academic Integrity

- Your code must be your own. You may use AI assistants (Claude, ChatGPT, Copilot) as learning tools, but you must understand and be able to explain every line you submit.
- If you use code from external sources, cite it explicitly.
- In your video, you must explain your own code. If you cannot explain it, or if your explanations are inconsistent with your understanding, it will be treated as plagiarism.
- The instructor reserves the right to request a short live clarification (in person or online) if the video leaves any doubt about authorship or understanding.
- Both team members are equally responsible for the entire project.

Build something you're proud of. Not something you copied.